
Do Off-the-Shelf Detector Embeddings Reveal Bowhead Whale Call Sub-Types? A Probing and Clustering Study on Weakly-Labeled Passive Acoustic Data

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Abstract

Manual bioacoustic taxonomies frequently include a catch-all category for sounds that resist clean classification; for migrating bowhead whales (*Balaena mysticetus*), roughly a third of manually annotated calls across seven morphological "Type" labels fall into categories described by human analysts as ambiguous or context-dependent. A companion acoustics paper establishes a convolutional detector that separates bowhead calls from other transient sounds with high accuracy (ROC-AUC ≈ 0.96 on a fully independent, 199,825-sample generalization test), but that detector is explicitly binary and does not model call sub-type structure. Here we ask two narrower questions about the representations available from that detector's development: (1) how much information about the seven manually annotated call sub-types is linearly decodable from embeddings trained only for reconstruction (a convolutional autoencoder) or for binary call/non-call detection (from-scratch and autoencoder-warm-started classifiers); and (2) whether unsupervised clustering of these same embeddings recovers the manual sub-type taxonomy without using any sub-type labels. We find that a linear probe recovers moderate sub-type signal (best macro one-vs-rest ROC-AUC 0.663, top-1 accuracy 36.5% against a 14.3% random baseline over seven classes), and that embeddings retaining more reconstructive detail (the autoencoder and warm-started classifier) are noticeably more informative for sub-type identity than embeddings optimized purely for binary detection (scratch classifier: macro ROC-AUC 0.577). In contrast, standard unsupervised clustering (k-means, BIC-selected Gaussian mixtures) fails to recover the manual taxonomy in any embedding space we tested (adjusted Rand index ≤ 0.055 in every configuration), indicating that whatever call-type signal exists in these embeddings is not axis-aligned with simple cluster geometry. We argue this dissociation – weak-to-moderate linearly decodable structure alongside a near-total failure of unsupervised clustering – is itself the useful empirical finding: it shows that representations optimized for reconstruction or binary detection are an insufficient basis for the unsupervised call-type subdivision envisioned in prior work, and motivates representation-learning objectives (e.g., contrastive or metric-learning pretraining) explicitly targeted at sub-type structure, which we outline as the next step in this research program.

1 Introduction

Between 2007 and 2014, 35 passive acoustic recorders (Directional Autonomous Seafloor Acoustic Recorders, DASARs) were deployed along a 280 km stretch of the Alaskan North Slope to monitor the fall migration of the Bering–Chukchi–Beaufort bowhead whale population. Manual analysts

annotated millions of detected calls with a time, frequency range, and one of seven morphological "Type" labels. A long-standing observation in this literature, reflected directly in the manual annotation handbook used to build this dataset, is that a substantial fraction of calls resist clean categorization into simple frequency-modulated archetypes (up-calls, down-calls, undulations) and are instead placed into broader, harder-to-define categories describing mixed frequency/amplitude modulation with broadband energy. Whether this catch-all grouping reflects a single underlying vocalization type or several distinct call types that happen to be difficult for a human analyst to separate by eye has been an open question for at least two decades, and is precisely the question a purely supervised classifier – which can only reproduce the labels it is given – cannot answer on its own.

A companion acoustics paper describes a two-stage detector for this dataset: a convolutional autoencoder that compresses single-channel SNR spectrograms into a 32-dimensional latent space, and a supervised classifier (trained either from random initialization, or warm-started from the autoencoder's encoder weights) that distinguishes bowhead calls from other transient sounds, including seismic airgun pulses, with ROC-AUC ≈ 0.96 on a fully independent held-out evaluation set. That detector is deliberately binary, and treats all seven manually annotated call sub-types identically during training and evaluation. This raises a natural follow-up question that the detector itself was not designed to answer: *do the representations produced along the way to building that binary detector already carry usable information about call sub-type, and if so, is that information organized in a way that unsupervised clustering could exploit to automatically discover or refine sub-type structure without additional manual annotation?*

We treat this as an empirical probing question rather than assume an answer. Concretely, we study three 32-dimensional embedding spaces that already existed as artifacts of building the binary detector – a purely reconstructive autoencoder embedding, a from-scratch binary-detection embedding, and an autoencoder-warm-started binary-detection embedding – and ask (RQ1) how much of the manually annotated seven-way call-type structure is linearly decodable from each, and (RQ2) whether standard unsupervised clustering methods (k-means, Gaussian mixtures with BIC-selected component count) recover that same seven-way structure without being given any sub-type labels. Our contribution is not a new architecture, but a carefully controlled empirical answer, using a leakage-free evaluation protocol inherited directly from the companion detector paper, to a question that is foundational to any future attempt at unsupervised or weakly-supervised bowhead call-type discovery: representations trained for reconstruction or binary detection are demonstrably *not* sufficient, on their own, for that goal, which motivates the representation-learning objectives we outline as future work in Section 6.

2 Related work

Bioacoustic deep learning. Convolutional and, increasingly, transformer-based architectures have substantially improved automated detection of marine mammal and avian vocalizations over classical spectrogram-correlation and hand-engineered feature approaches [2, 3]. Prior bowhead-specific work used hand-crafted features and a shallow three-layer network to flag calls without attempting sub-type classification [1]. Our companion detector paper replaces this with a modern convolutional detector, but likewise stops short of sub-type classification, which is the gap this paper addresses.

Representation learning for bioacoustics. Convolutional autoencoders and related unsupervised encoders have been used to compress bioacoustic spectrograms into low-dimensional embeddings for downstream clustering and visualization [4, 5]. Most such studies, however, evaluate reconstruction quality or binary/coarse detection performance rather than directly asking whether the resulting embedding geometry supports *unsupervised* recovery of a pre-existing fine-grained taxonomy, which is the specific question we address here.

Weakly-labeled and open-set classification. Because the "complex call" category in this dataset was defined by human analysts as a catch-all for signals that did not fit cleaner categories, it is best understood as a weak or noisy label rather than a well-defined class. This connects our study to the broader literature on learning under label noise and open-set/novel-class discovery, where a central question is whether representations trained for a related but distinct task (here, binary detection) transfer usefully to separate finer-grained or entirely novel classes not explicitly supervised during training.

Table 1: Manually annotated call-type counts in the grouped train/test split used for the probing and clustering study (identical split convention as the companion detector paper).

	Type 1	Type 2	Type 3	Type 4	Type 5	Type 6	Type 7	Total
Train	13,951	10,049	7,576	4,548	3,292	1,456	6,528	47,400
Test	742	701	392	215	165	97	288	2,600

3 Data and problem setup

We reuse, without modification, the leakage-controlled 100,000-sample dataset and grouped date×site train/test split described in the companion detector paper: 50,000 manually verified bow-head calls (Types 1–7) and 50,000 auto-detected non-call transients (Type 0), drawn from DASAR sites 3 and 5 and DASARs A, D, and G. Every image is a 121×104 single-channel SNR spectrogram spanning 3 seconds and 25–500 Hz, centered on the time of peak local power spectral density (see companion paper for full preprocessing details). Splitting is grouped by the date×site key used to record each detection, so that all images from a given site on a given day fall entirely within one partition – the same leakage-safety invariant used throughout the companion paper – yielding 91,920 training images (43 groups) and 8,080 held-out test images (8 groups), with zero group overlap between partitions.

For the present study we restrict attention to the 50,000 manually verified calls, which carry one of seven Type labels (Table 1). We treat these Type labels exactly as delivered by the original manual-analysis pipeline: as opaque integer categories, with no semantic mapping (e.g. "up-call," "complex call") reconstructed or assumed, since no such mapping exists in the dataset artifacts available to us. This is a deliberate choice: any claim about which Types correspond to "simple" or "complex" calls in the sense used by the original annotation handbook would require information we do not have, so we evaluate all seven Types identically and let the data speak for itself. Restricting to the grouped split above yields 47,400 training and 2,600 test calls with Type labels (Table 1); Type 6 is the rarest (1,553 examples total) and Type 1 the most common (14,693 examples total), a roughly 9.5:1 imbalance ratio that we account for using class-balanced probe training (Section 4).

4 Methods

Embedding spaces. We evaluate three 32-dimensional embedding spaces, all already produced during the construction of the companion binary detector, with no additional training performed for this study:

- **Autoencoder (AE):** the encoder trunk of the convolutional autoencoder, trained purely to minimize pixel-wise reconstruction error, with no access to any call/non-call or Type label during training.
- **CNN (scratch):** the encoder trunk of the binary call/non-call classifier, trained end-to-end from random initialization using only the binary label.
- **CNN (warm-start):** the same binary classifier architecture, but with the encoder trunk initialized from the trained autoencoder and then fine-tuned end-to-end on the binary label.

All three share an identical architecture (three Conv+BatchNorm+ReLU+MaxPool blocks, FC-64→FC-32 bottleneck), differing only in what objective (if any) supervised their training, isolating the effect of training objective on call-type informativeness while holding architecture and input data fixed.

Linear probing (RQ1). For each embedding space, we extract 32-dimensional embeddings for every manually-verified call in the grouped train/test split (Section 3) and fit a class-balanced, multinomial logistic-regression probe (StandardScaler → balanced multinomial logistic regression, L_2 regularization $C = 1$) on the training embeddings and Type labels, following standard frozen-embedding probing methodology. We report one-vs-rest ROC-AUC per Type, macro-averaged ROC-AUC across the seven types, and top-1 accuracy, all computed on the held-out grouped test partition. A macro ROC-AUC of 0.5 and a top-1 accuracy of $1/7 \approx 14.3\%$ correspond to chance performance under this seven-way task.

Table 2: Seven-way call-type linear-probe results (macro one-vs-rest ROC-AUC, top-1 accuracy, and per-Type ROC-AUC) on the held-out grouped test split (chance: macro ROC-AUC 0.5, top-1 accuracy 14.3%).

Embedding	Macro ROC-AUC	Top-1 Acc.	Per-type ROC-AUC						
			T1	T2	T3	T4	T5	T6	T7
Autoencoder	0.659	36.1%	0.735	0.722	0.564	0.619	0.596	0.584	0.794
CNN (scratch)	0.577	15.9%	0.606	0.573	0.542	0.570	0.520	0.583	0.645
CNN (warm-start)	0.663	36.5%	0.744	0.754	0.580	0.623	0.579	0.604	0.754

Unsupervised clustering (RQ2). For each embedding space, we standard-scale the pooled train+test call embeddings (50,000 points) and apply two clustering algorithms: (i) k-means with $k = 7$, matching the number of manually annotated Types, and (ii) a Gaussian mixture model with the number of components selected by Bayesian Information Criterion (BIC) over $k \in \{2, 4, 6, 7, 8, 10, 12, 16\}$, allowing the data to select a different granularity than the manual taxonomy if warranted. We score cluster quality using the silhouette coefficient (an unsupervised, label-free measure of cluster separation) and, critically, score *agreement with the manual Type labels* using the adjusted Rand index (ARI) and normalized mutual information (NMI) – two standard clustering-agreement metrics that are 0 in expectation for a random partition and 1 for a perfect match to the reference labels.

5 Results

5.1 RQ1: call-type information is linearly decodable, but modest

Table 2 reports the linear-probe results for all three embedding spaces. All three exceed the chance baseline (macro ROC-AUC 0.5, top-1 accuracy 14.3%) by a wide margin, confirming that call-sub-type information is present and linearly accessible in these embeddings despite none of them being trained with sub-type supervision. However, the gap between embedding spaces is substantial and systematic: the autoencoder and warm-started embeddings (macro ROC-AUC 0.659 and 0.663) are both markedly more informative for call sub-type than the from-scratch binary-detection embedding (macro ROC-AUC 0.577). This pattern holds for six of seven individual Types (Table 2), and is consistent with the interpretation that an embedding optimized purely to separate "call" from "non-call" discards within-call visual detail that is not needed for that binary decision, but that a reconstructive or partially-reconstructive objective retains. Types 1, 2, and 7 are the most separable across all three embeddings (per-type ROC-AUC 0.72–0.79 for AE/warm-start); Types 3, 5, and 6 are consistently the hardest to separate (ROC-AUC 0.52–0.60), independent of which embedding is used.

5.2 RQ2: unsupervised clustering fails to recover the manual taxonomy

Table 3 reports clustering results for all three embeddings. In every configuration – both algorithms, all three embedding spaces – agreement with the manual seven-way Type labels is close to zero ($\text{ARI} \leq 0.055$, $\text{NMI} \leq 0.055$), far below what would indicate even a loose correspondence between discovered clusters and the manual taxonomy. This holds despite k-means finding geometrically well-separated clusters in some embeddings by its own internal criterion (silhouette 0.306 for the scratch-CNN embedding) – meaning the clusters that do exist in these embedding spaces are organized around structure other than manually annotated call type (plausibly SNR, site, DASAR channel, or signal duration, none of which were controlled for in this analysis). The BIC criterion for the Gaussian mixture model selected a substantially larger number of components (12–16) than the seven manually annotated types in every embedding, and even at that finer granularity, agreement with the manual Types did not improve materially.

6 Discussion and limitations

Taken together, these results paint a consistent picture: call-type information exists in all three embedding spaces at a level well above chance (RQ1), but is not organized in a way that any of the unsupervised clustering methods we tested can recover (RQ2). We consider this dissociation – rather

Table 3: Unsupervised clustering agreement with the manual seven-Type taxonomy, pooled across the full 50,000 manually-verified calls. Silhouette is a label-free internal cluster-quality score; ARI and NMI measure agreement with the manual Type labels (0 = chance, 1 = perfect).

Embedding	Method	k	Silhouette	ARI	NMI
Autoencoder	k-means	7	0.102	0.015	0.031
Autoencoder	GMM (BIC)	16	-0.020	0.031	0.055
CNN (scratch)	k-means	7	0.306	0.008	0.016
CNN (scratch)	GMM (BIC)	12	-0.136	0.010	0.020
CNN (warm-start)	k-means	7	0.044	0.016	0.024
CNN (warm-start)	GMM (BIC)	16	-0.088	0.012	0.022

than either result in isolation – to be the main finding of this study. It shows that a decodable linear direction for call-type is not sufficient for an unsupervised clustering algorithm to discover it unprompted; the relevant information may be encoded in a low-variance or non-axis-aligned subspace that reconstruction- or detection-oriented objectives have no incentive to make geometrically salient. Concretely, this means the unsupervised subdivision of the "complex call" catch-all category envisioned in prior work cannot be accomplished by naively clustering an embedding trained for a different objective (reconstruction or binary detection); it requires an embedding explicitly optimized to make call-type structure clusterable, motivating the contrastive- or metric-learning-based objectives we outline below.

We also note a second consistent pattern that is scientifically interesting in its own right: the from-scratch binary-detection embedding is uniformly worse at call-type discrimination than either the autoencoder or the warm-started embedding, despite performing marginally *better* than the warm-started model at the binary detection task itself (per the companion detector paper). This suggests that optimizing purely for call/non-call separability actively discards information that is useful for finer-grained discrimination – a representation-learning trade-off that, to our knowledge, has not been explicitly quantified for this dataset before, and that has a direct practical implication: a detector optimized purely for binary recall should not be assumed to produce embeddings suitable for downstream sub-type analysis, even though it is architecturally identical to one that is.

Limitations. This study has several limitations that qualify how strongly its conclusions should be read. First, we treat the manually annotated seven-way Type taxonomy as ground truth for the ARI/NMI clustering-agreement metrics, but the taxonomy’s own documentation describes at least one of its categories as an explicit catch-all for signals that resisted clean human categorization; a low ARI could therefore reflect either an uninformative embedding or a manual taxonomy that does not itself correspond to a naturally clusterable structure – our data cannot distinguish between these two explanations, and doing so is an important direction for future work (e.g., via targeted expert review of specific discovered clusters). Second, we evaluate only two clustering families (k-means, Gaussian mixtures) at a single random seed each; more flexible density-based methods (e.g., HDB-SCAN) or repeated-seed stability analysis might recover structure that these two methods miss, and we plan to report this in a full version of this study. Third, our linear probe is a single, comparatively simple classifier (multinomial logistic regression); a nonlinear probe could reveal additional decodable structure not visible to a linear decision boundary, which would strengthen rather than weaken the RQ1/RQ2 dissociation reported here. Fourth, none of the three embeddings evaluated here were trained with any objective that specifically targets sub-type structure (e.g., contrastive learning with call-preserving augmentations, or metric learning against partial sub-type supervision); this is precisely the gap our findings motivate addressing next.

Future work. The most direct next step is to replace the purely reconstructive or purely binary-discriminative pretraining objectives studied here with a representation-learning objective explicitly designed to make call-type structure separable – for instance, a contrastive objective using label-preserving spectrogram augmentations (time-shift, additive noise at controlled SNR, frequency-axis jitter) to pull same-call views together and push different-call views apart, or a semi-supervised metric-learning approach that uses a small fraction of Type-labeled calls to shape the embedding geometry before applying unsupervised clustering to the remainder. Any such method should be validated not only by ARI/NMI against the existing manual taxonomy, but by a human-expert review of discovered clusters that do *not* match the manual taxonomy, since – as noted above – the

manual taxonomy’s own "complex call" category was defined as a catch-all rather than a settled ground truth, and a genuinely useful method might legitimately propose a different partition than the one used here for evaluation. Beyond call-type structure, the original research program motivating this dataset also identified single-spectrogram range estimation and multi-sensor context fusion (combining detections across co-located DASAR channels, and explicitly modeling seismic airgun temporal periodicity to reduce false alarms) as further open objectives; neither is addressed by the present study or its companion detector paper, and both remain open targets for future work building on the representations studied here.

Broader impact

Automated tools of this kind are intended to help researchers process existing, currently under-analyzed decades-scale passive acoustic archives more efficiently, in support of monitoring bowhead whale distribution and behavior amid climate change and increasing human activity in the Arctic – a topic of direct importance to Alaska Native communities who rely on the bowhead subsistence hunt. We are not aware of a plausible negative-use pathway for this specific probing/clustering study, which does not itself produce a deployable detector or classifier.

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